

4(a)	force per unit positive charge	B1
4(b)(i)	1 $E = V/d$ or $E = \Delta V / \Delta d$ $d = 4.0 \times 10^3 / 5.0 \times 10^4$	C1
	$= 8.0 \times 10^{-2} \text{ m}$	A1
	2 plates are (in) horizontal (plane) (above and below the rod)	B1
	top (plate) negative and bottom (plate) positive	B1
4(b)(ii)	magnitude $= 5.0 \times 10^4 \times 3 \times 1.6 \times 10^{-19}$ $= 2.4 \times 10^{-14} \text{ N}$	A1
	direction is (vertically) downwards / down	B1

Question	Answer	Marks
4(b)(iii)	$6.2 \times 10^{-16} = 2.4 \times 10^{-14} \times 72 \times 10^{-3} \times \cos \theta$	C1
	$\theta = 69^\circ$	A1

Question	Answer	Marks
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